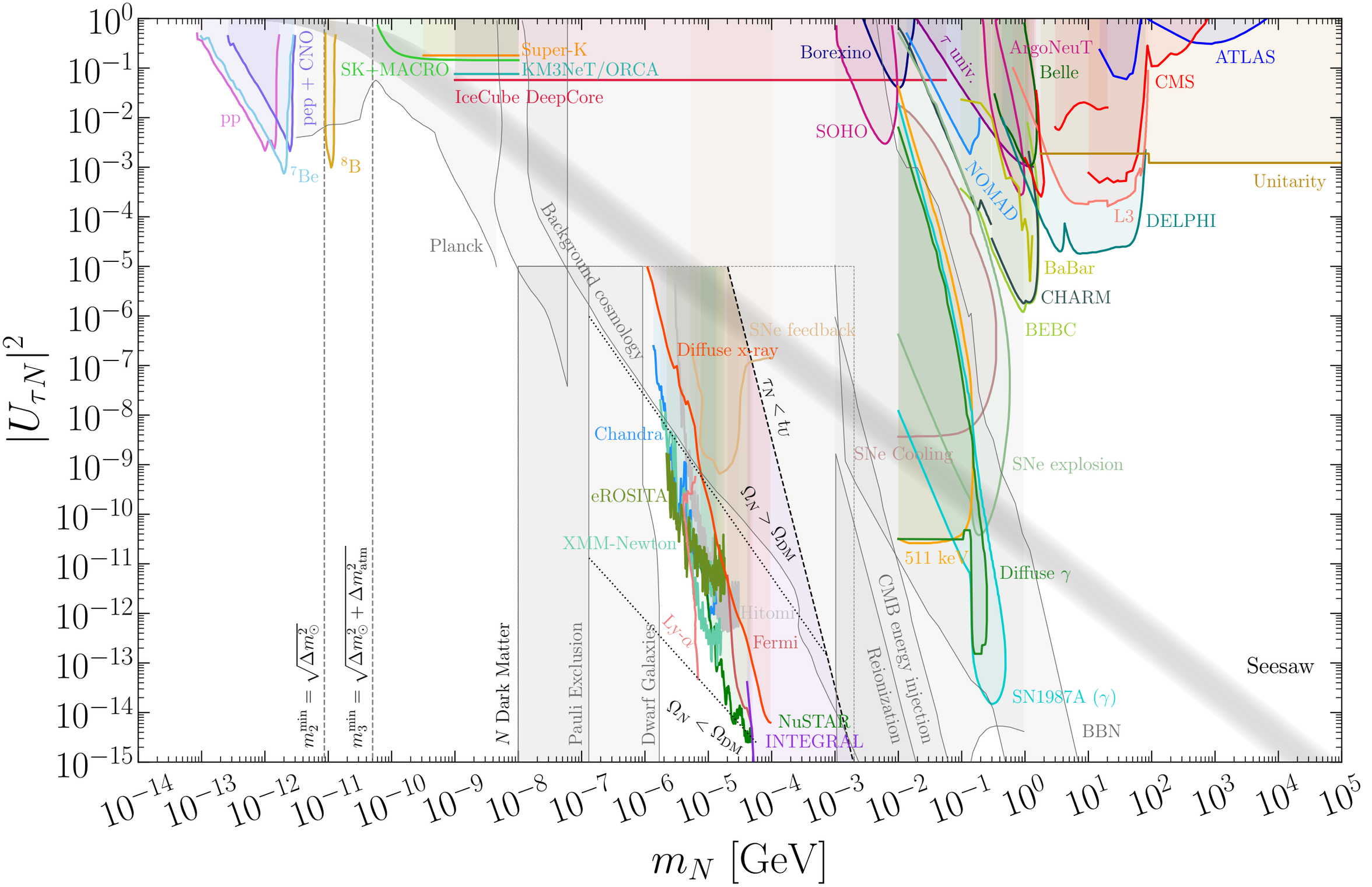

$|U_{\tau N}|^2$
m_N [GeV]
pp
pep + CNO
^{7}Be
^{8}B
SK+MACRO
Super-K
KM3NeT/ORCA
IceCube DeepCore
Borexino
τ univ.
ArgoNeuT
Belle
ATLAS
CMS
SOHO
NOMAD
Unitarity
L3
DELPHI
BaBar
CHARM
BEBC
Planck
Background cosmology
SNe feedback
Diffuse x-ray
$\tau_N < t_U$
Chandra
SNe Cooling
SNe explosion
eROSITA
$\Omega_N > \Omega_{\rm DM}$
XMM-Newton
511 keV
Diffuse γ
Hitomi
Ly-α
Fermi
$\Omega_N < \Omega_{\rm DM}$
NuSTAR
INTEGRAL
SN1987A (γ)
BBN
Seesaw
N Dark Matter
Pauli Exclusion
Dwarf Galaxies
CMB energy injection
Reionization
$m_2^{\rm min} = \sqrt{\Delta m_\odot^2}$
$m_3^{\rm min} = \sqrt{\Delta m_\odot^2 + \Delta m_{\rm atm}^2}$